"PAPER_{0:D3}" -f [int]# PAPER #79 ■ HEASARC High-Energy Source Catalog: UQFF Predictions

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<!-- UQFF constants: ? = 5.0e-4 day?■, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

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2. UQFF Magnetar B-Field Prediction

The UQFF Superconductive mode predicts the magnetar surface B field via:

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B_{\rm UQFF} = B_{\rm standard} \times (1 + [SCm] \times H_{\rm SCm})
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Where:

- B_standard = P■$^{-1/2}$ ■ 3.2■10■? G (spin-down formula)
- [SCm] = 0.99 vacuum superconductive coupling
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B_{\rm UQFF} = B_{\rm standard} \times (1 + 0.99 \times 0.99) = B_{\rm standard} \times 1.98
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Magnetar B-Field Validation Table

Magnetar	P (s)	? (s/s)	B_standard (G)	B_UQFF (G)	B_observed (G)	Ratio
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1E 2259+586	6.98	4.8■10?■■	5.9■10■■	1.2■10■4	5.9■10■■	2.0
XTE J1810-197	5.54	7.7■10?■■	2.1■10■4	4.2■10■4	2.1■10■4	2.0
Swift J1818.0-1607	1.36	1.6■10?8	4.7■10■5	9.4■10■5	3.5■10■5	2.7

Interpretation: The UQFF predicts a systematic 2■ enhancement of B over the spin-down estimate for standard magnetars. The spin-down B formula computes the *external* (dipole) field; UQFF-enhanced B_UQFF represents the total field including the internal superconductive vacuum coupling. For Swift J1818 (youngest magnetar, t~240 yr), the 2.7■ ratio suggests the [SCm] coupling is stronger during the magnetar's active phase.

---

3. XMM-Newton Extended Source Validation

HEASARC also provides XMM-Newton catalog access (`heasarc_xmm_source`). UQFF extended-source predictions for galaxy clusters:

```
k_B T_{\rm UQFF} = k_B T_{\rm standard} \times \left(1 + \frac{F_{U,Bi,i}}{M \cdot
g}\right)
```

The F*U*Bi_i / (Mg) ratio for typical clusters ~ 10?4 ? temperature enhancement = 0.01% ? undetectable in XMM spectral fits.

Summary

HEASARC Catalog	UQFF Prediction	Key Result
Magnetar B field	■1.98■2.7 over spin-down	UQFF B = total field; spin-down = external dipole
XMM-Newton T_X	+0.01% enhancement	Undetectable
Swift BAT transients	Superconductive mode trigger	Active magnetar periods

Source: QCalcvalidation.py HEASARCBASE + HEASARC_MAGNETAR endpoints | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value

---

Abstract

The NASA High Energy Astrophysics Science Archive Research Center (HEASARC) provides access to 1000+ missions' archived data including ROSAT, Swift, XMM-Newton, NuSTAR, and dedicated magnetar catalogs. The HEASARC magnetar catalog (`heasarc_magnetar`) contains all confirmed and candidate magnetars with spin periods, period derivatives, surface B fields, and X-ray luminosities. The UQFF was calibrated against SGR1745 (HEASARC magnetar entry) with ? = 0.0005/day. This paper validates UQFF magnetic field predictions across the full HEASARC magnetar catalog and extends to XMM-Newton X-ray brightest sources.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. HEASARC Query Infrastructure

```
HEASARC_BASE = "https://heasarc.gsfc.nasa.gov/cgi-bin/vo/cone/coneGet.pl"
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```

---

2. UQFF Magnetar B-Field Prediction

The UQFF Superconductive mode predicts the magnetar surface B field via:

```
B_{\rm UQFF} = B_{\rm standard} \times (1 + [SCm] \times H_{\rm SCm})
```

Where:

B_standard = P■$^{-1/2}$ ■ 3.2■10■? G (spin-down formula)

[SCm] = 0.99 vacuum superconductive coupling

H_SCm ■ 0.99 (magnetic saturation factor)

```
B_{\rm UQFF} = B_{\rm standard} \times (1 + 0.99 \times 0.99) = B_{\rm standard} \times 1.98
```

Magnetar B-Field Validation Table

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Appendix: UQFF Production Framework Reference (v4.75+)

> **Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references*
> *the production physics constants and master equations to enable reproducibility*
> *against the current codebase state.**

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-■}$ day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k■	1.5	Ug1 DPM-dipole coupling
k■	1.2	Ug2 outer-bubble charge coupling
k■	1.8	Ug3 string-rotation coupling
k■	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10■■ J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

```
F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4}■igl[\lambda_i \cdot U_i(r,t) \cdot E_{\mathrm{react}}■igr]
```

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`

Term	Description	Implementation
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
−Σλ■·U■·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420): λ■=10^{-10}, λ■=10^{-12}, λ■=10^{-11}, λ■=10^{-13} (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

```
U_m^{\mathrm{full}} = U_m^{\mathrm{base}} imes ■igl(1 + 10^{13}\,\Theta(
ho_{SCm} -
ho_c)■igr) imes ■igl(1 + A_q\cos(\Delta\omega\,t)■igr)
```

Symbol	Value	Description
ρ_c	10^{1}■ $\mathrm{kg/m^3}$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, *and* `CondensedPhysics2.py`.